

# Uplift / CATE Performance Summary (Track R — Research-generated IncentiveRewards mock)

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*Milestone 6 — grouped multi-seed evaluation*

**TreatmentSourceBasis = Research-generated IncentiveRewards mock. Milestone 7 must not consume these scores as production lift.**

## Sample

| Metric               | Value     |
|----------------------|-----------|
| Rows                 | 11669     |
| Members              | 6668      |
| Treated rows         | 5217      |
| Control rows         | 6452      |
| Treated closure rate | 0.0219    |
| Control closure rate | 0.0225    |
| ESS ratio            | 0.9990    |
| CATE identified      | Yes       |
| Treatment grain      | gap-level |

## Per-seed holdout

| Seed | Ate    | Auuc    | Placebo Auuc |
|------|--------|---------|--------------|
| 42   | 0.0606 | -0.0042 | 0.0011       |
| 123  | 0.0141 | 0.0034  | 0.0008604    |
| 456  | 0.0177 | -0.0016 | -0.0018      |
| 789  | 0.0255 | -0.0024 | 3.683e-05    |
| 2026 | 0.0463 | -0.0029 | -0.0009375   |

## Care-gap descriptives (Track R mock assignment)

T is Bernoulli among gaps still open at reconstructed outreach. Y is 90-day post-issue InComm closure, not the raw Closed flag. Treatment and control should both have outcome variation; T must not equal Y.

| Care Gap Code | N | Treated Rows | Control Rows | Treated Closure Rate | Control Closure Rate | Untreated Arm Empty |
|---------------|---|--------------|--------------|----------------------|----------------------|---------------------|
|---------------|---|--------------|--------------|----------------------|----------------------|---------------------|

|                             |      |     |     |        |        |    |
|-----------------------------|------|-----|-----|--------|--------|----|
| <b>BH_POST_DISCHARGE_7D</b> | 1667 | 729 | 938 | 0.0041 | 0.0043 | No |
| <b>COPD_SPIROMETRY</b>      | 1667 | 762 | 905 | 0      | 0.0011 | No |
| <b>DIAB_A1C_TEST</b>        | 1667 | 752 | 915 | 0      | 0.0011 | No |
| <b>DIAB_BLOOD_SUGAR</b>     | 1667 | 735 | 932 | 0.0014 | 0.0011 | No |
| <b>DIAB_EYE_EXAM</b>        | 1667 | 754 | 913 | 0.0013 | 0.0011 | No |
| <b>DIAB_NEPHRO_SCREEN</b>   | 1667 | 726 | 941 | 0      | 0.0011 | No |
| <b>HTN_PCP_FOLLOWUP</b>     | 1667 | 759 | 908 | 0.1436 | 0.1498 | No |

### Why diabetes looked untreated-empty (member-broadcast T)

M4 rolled sim rewards to MemberId and copied T onto every gap. A closed BH reward then marked still-open diabetes gaps as treated.

| Care Gap Code               | N    | Treated Rows | Control Rows | Treated Closure Rate | Control Closure Rate | Untreated Arm Empty |
|-----------------------------|------|--------------|--------------|----------------------|----------------------|---------------------|
| <b>BH_POST_DISCHARGE_7D</b> | 1667 | 1248         | 419          | 1.0000               | 0                    | No                  |
| <b>COPD_SPIROMETRY</b>      | 1667 | 1011         | 656          | 1.0000               | 0                    | No                  |
| <b>DIAB_A1C_TEST</b>        | 1667 | 1667         | 0            | 0.1980               | —                    | Yes                 |
| <b>DIAB_BLOOD_SUGAR</b>     | 1667 | 1667         | 0            | 0.3185               | —                    | Yes                 |
| <b>DIAB_EYE_EXAM</b>        | 1667 | 1667         | 0            | 0.2915               | —                    | Yes                 |
| <b>DIAB_NEPHRO_SCREEN</b>   | 1667 | 1667         | 0            | 0.2987               | —                    | Yes                 |
| <b>HTN_PCP_FOLLOWUP</b>     | 1667 | 622          | 1045         | 1.0000               | 0                    | No                  |

## Identification autopsy

- IncentiveRewards 3rd-party feed is not implemented (InComm 2026-07-20 Q4). Track R uses a research-generated IncentiveRewards-shaped ledger on real members/gaps/amounts. It is not InComm-issued and must not feed Milestone 7.
- Live QA: IsRewardEligible iff Status='Closed' (0 violations). HAR cases / sim rows are generated for those closed gaps (CaseSubtype = care\_gap\_code). Gap-level Track S T equals Y on catalog gaps.
- Member-broadcast T (M4 lifetime sim count) copies any reward onto every gap for that member. Diabetes untreated-empty is that artifact, not a missing arm.
- ISS-008: clinical claims ServiceFrom  $\leq$  2026-06-23; HAR CreatedAtUtc  $\geq$  2026-06-25. Case time is after every claim on this extract — reverse causation is structural for HAR. The Track R mock reconstructs outreach lag from gap tenure.
- Track R mock assignment is Bernoulli among gaps still open at t0. Historical closure was not caused by mock T; identified CATE here is a pipeline/negative-control result, not partner-facing lift.



